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Volumetric Crop Yield

Project Specifications & Risk Assessment

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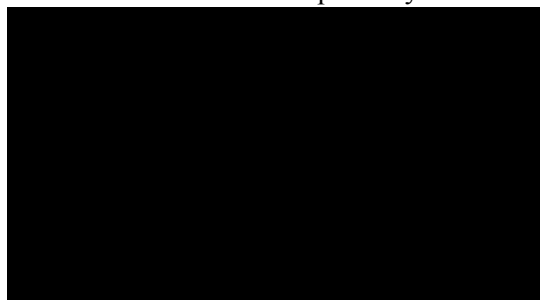


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1. High-Level Project Description

1.1. Motivation

Agricultural productivity measures the ratio of agricultural outputs to agricultural inputs [1]. There are many factors that influence agricultural productivity, these include the use of fertilizers, irrigation, herbicides, pesticides and crop planning [2]. In order to optimize these input parameters, an understanding of the respective output parameter must be developed: “yield.” Yield is defined as the crop output per unit of land [3].

The rapid analysis of agricultural yield of all types of crops is crucial to the efficient and competitive operation of farmers within those industries [1]. It is of great importance for farmers to be able to assess the state of their crops, understand the distribution of yield over the field, and make judgements on the quality of their harvest in relation to their farming practices and local environmental factors. Overall, every farm is different and many farmers lack the insight they need to adequately optimize their use of resources. This information can be used for crop insurance purposes, harvest delivery estimates, planning storage requirements, and budgeting [4]. This is especially important for small to mid-sized farmers who cannot afford a great amount of labour or expensive equipment.

Existing yield systems are weight based and are not capable of detecting items with low weights relative to the weight of the conveyor system in use. Moreover, such systems can become very expensive and are constrained by the conveyor system the farming facility has already adopted [5]. Systems used in factory automation settings can potentially be adopted; however, they are excessively expensive and, moreover, not suitable for outdoor operation in potentially dusty conditions [6]. Farmers desire a system that addresses these concerns while also maintaining a standard of accuracy and robustness.

1.2. Project Objective

The objective is to design a low-cost integrated system that is capable of detecting the volume of an agricultural crop passing by on a conveyor belt during the harvesting process and upload the results to a web portal for reporting purposes.

1.3. System Block Diagram

The high level design of this volumetric crop sensing system is outlined in Figure 1-1. Overall, five core subsystems are to be designed for power control, processing, optical sensing, data services, and data visualization. Each of these subsystems and their respective intent are described in Table 1-1.

Table 1-1. A high level description of each core subsystem, its intended purpose, and how it interfaces with other subsystems and subcomponents.

Subsystem	Description
Power Control & Distribution	Ensures that power is properly regulated and distributed to each of the hardware subsystems in such a way that matches each component's specifications/requirements.
Sensing Platform	The sensing platform consists of a sensor array that exists over the conveyer belt. Due to the variable light conditions of the operating environment, this platform will also host a lighting fixture that will ensure an adequate amount of light is present for the sensors to operate.
Processing Platform	The central component that aggregates the data provided by the sensor subsystem with dynamic timing and GPS information to be locally cached. Once enough data has been consolidated and a web connection is present, this system interfaces with a data input web service to off-load the necessary data for further processing.
Data Warehousing & Web Services	This platform is intended to aggregate the data from all the modules present in the field and store the data in a database server for on-demand access from the client. This system also performs data validation and discards/corrects any erroneous data (e.x. Impractical GPS coordinates, outlier data points, etc.). The platform also mines the data and generates reports that are exposed to the client data visualization applications.
Client Data Visualization Applications	A web based mobile application and accompanying website that presents the reports generated by the backend services in an accessible and easy to read manner.

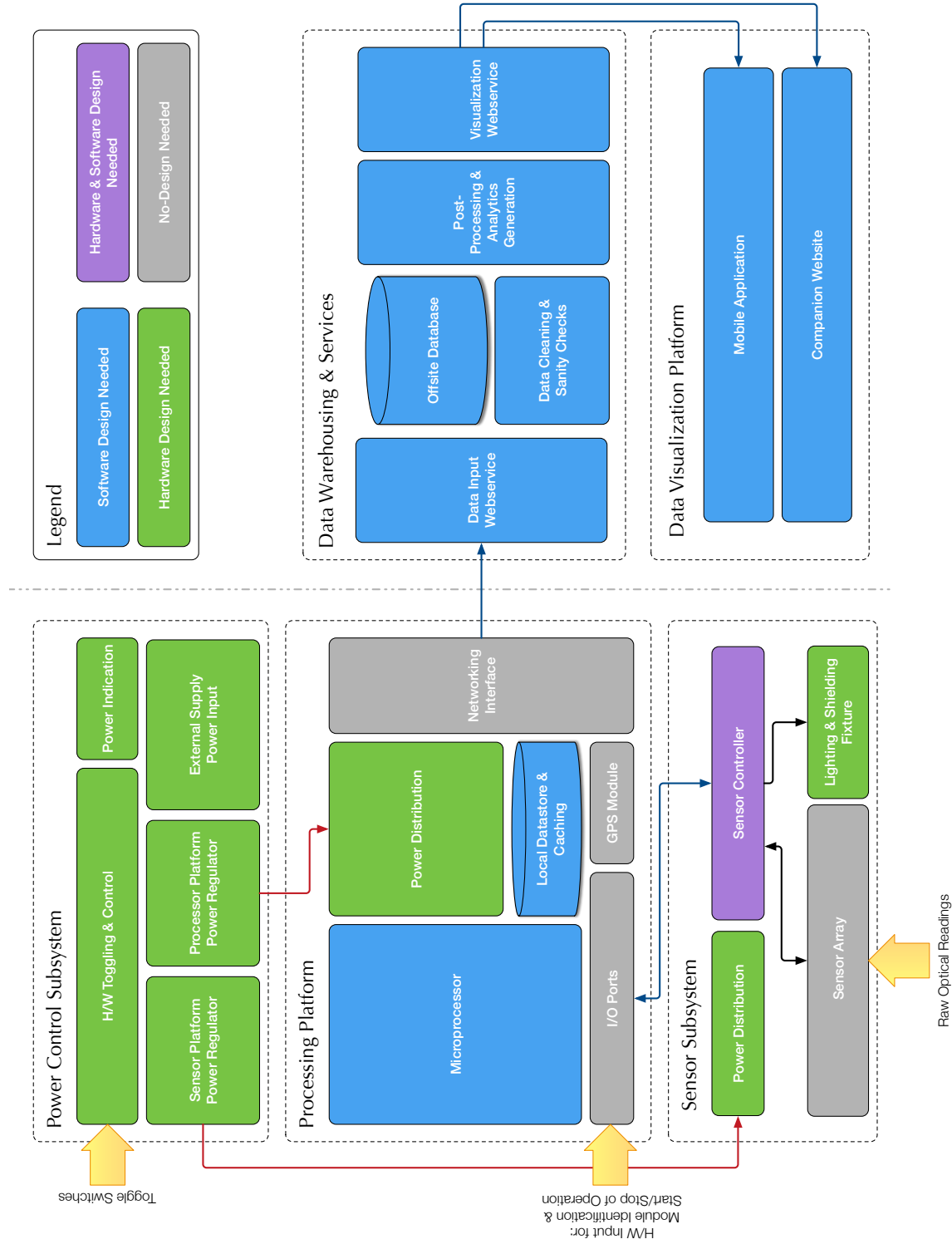


Figure 1-1. Block diagram overview of the intended system and the various subcomponents.

2. Project Specifications

The project specifications are divided into two groups: functional specifications and non-functional specifications. Table 2-1 and Table 2-2 respectively outline the functional and non-functional specifications for the intended system.

Table 2-1. Functional specifications of the system.

Specification	Subsystem	Necessity	Description
Power Requirement	Power Control Subsystem	Essential	The typical tractor the system is intended to be installed on will provide a supply voltage of 12 V [7].
Crop Yield Accuracy	Processing Platform	Essential	Must be able to calculate volumetric yield during the harvesting process with an error of less than 15% from the actual.
Data Storage	Data Warehousing & Services	Non-Essential	Must have the ability to transfer and store the volumetric data produced from the system to an off-site computing platform for archiving.
Data Visualization	Data Visualization Platform	Non-Essential	Must provide an interface from which an operator can see a clear visualization of their crop yield over the geographic area of the harvested field.
Data Processing Rate	Processing Platform	Non-Essential	Must operate in real time; latency between sensor input and volumetric data output must be less than 2 seconds for conveyer speeds of 30 cm/s.
Image Pick-up	Sensor Subsystem	Essential	System must be able to acquire image in lighting conditions anywhere from 0 lux (complete darkness) to 100 000 lux (full daylight).
Yield Location	Processing Platform	Essential	Must be able to match yield data with geographic location data from the on-board GPS module.

Table 2-2. Non-functional specifications of the system.

Specification	Subsystem	Necessity	Description
Cost	All	Non-Essential	Must cost less than \$1500 for each hardware module.
Installation Complexity	All	Non-Essential	Must take less than 4 hours to set up and train customers to operate the system.
Weight	All	Non-Essential	Product must weigh less than 10 kg to ensure that it is easy to carry, configure, and install.
System Integration	All	Essential	Must be able to attach to variable sizes of conveyor belts between 0.91 m (3 ft) to 1.22 m (4 ft) wide.
Temperature Resistant	All	Essential	System should be able to operate at temperatures of -20°C to 40 °C, to accommodate varying crop types and farming conditions [8].
Water Resistant	All	Essential	Able to be submerged in up to 30 cm of water and maintain functionality.
Vibration and Shock Resistant	All	Essential	Must be able to withstand the forces created from a moving tractor.

3. Risk Assessment

There are many possible risks when working on a multi-tiered and complex project such as this. Table 3-1 identifies possible situations that pose a significant amount of risk to the success of the project.

Table 3-1. Various potential risks, probability of occurrence, their associated impact, and possible solutions to mitigate the risks.

1	Insufficient Experience in Image Processing
	Probability: Medium
	Impact: High
	Description: Most group members are unfamiliar with image processing and will be learning as the project progresses. This may delay the project or make some features and specifications unrealizable within the allotted timeframe.

	Mitigation Measure: There are many open-source resources available for image processing. The team should adapt existing libraries and processing algorithms whenever possible. If image processing is still too challenging and team members are unable to meet the specifications using this technique, other sensors can be used; for example, optical range finders. Such sensors would not require image processing; however, they do limit what can be done with the data.
2	System Complexity Impacting Timeline Probability: Medium Impact: High Description: This is a problem of both technical competence and time management. The team may run into the problem of not being able to complete all of the subsystems on time due to time constraints and the amount of research and work required. Mitigation Measure: We need to parallelize tasks as much as possible. Software can be done independently of the hardware in many cases. Because development of the algorithms and hardware can occur in parallel to some extent, we can work on the project over our co-op term to increase the amount of time we have available. If we discover that we are behind schedule, we can scale down the project to only meet core features and specifications. Data visualization can be removed and data warehousing systems can be scaled down if needed.
3	Imperfect Operating Environment May Hinder System Performance Probability: High Impact: High Description: Depending on what type of conveyor belt this system is used on, vibrations can prevent accurate sensor readings. This could result in blurry images and noisy signals from optical range finders. In addition, varying geometries and textures of different types of crop introduces exceptions and unexpected corner-cases for the processing platform. Insufficient lighting will introduce shadows and dark spots in gaps, preventing image processing algorithms from working properly. Mitigation Measure: To solve this problem, we can restrict the operating environment of the system by using conveyor belts that do not have large amounts of vibration. In addition, lighting fixtures can be used to improve the uniformity between measurements. If needed, the team can choose to optimize the software and hardware for specific cases that are the most desirable. In this scenario, the idea is to not implement a system that works on all crops; but, to have a system will work well for a specific subset of crops or crops with specific physical properties.

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